

Swathi Bhuvaneshwar Babu

PROFESSIONAL SUMMARY

- Senior Quality Assurance Lead with 14+ years of experience in test strategy, automation framework development, defect lifecycle management, and release quality across embedded, IoT, enterprise, and web platforms
- Deep expertise in JIRA for end-to-end QA workflow management — including custom workflow configuration, JQL-based reporting dashboards, sprint planning, backlog grooming, defect triage, fix-version tracking, and requirements traceability with XRay/Zephyr integration
- Proficient in Python-based test automation and framework development (Pytest, DeCAF, ACTS, RDK, Selenium) with hands-on experience designing, building, and maintaining scalable test suites for firmware, API, and UI validation
- Early adopter and power user of generative AI tools for QA engineering — using Kiro AI, Claude (Anthropic), Amazon Q Developer, and GitHub Copilot daily to generate test cases, automate code reviews, analyze failure logs, build dashboards, and produce release documentation at 3x speed
- Built AI-powered QA automation pipelines combining prompt engineering, agent orchestration (MCP, steering files, hooks), and RAG patterns — enabling autonomous retrospective generation, defect trend analysis, and wiki page creation directly from JIRA and Confluence data
- Proven ability to define, measure, and improve QA operational KPIs using JIRA analytics and custom dashboards — tracking defect escape rate, fix rate, detection latency, automation coverage, and release readiness metrics
- Experienced in validating embedded systems and TV platforms (FireTV, Roku, Google Nest) in RTOS/Linux environments — covering connectivity protocols (Bluetooth, Wi-Fi, Zigbee, Thread, Matter), system stability, and performance
- Strong expertise in system-level debugging using logs, telemetry, and ADB — performing deep root cause analysis and driving defect resolution through structured JIRA workflows and cross-team collaboration
- Championed responsible AI adoption within QA teams — evaluating LLM output accuracy, establishing prompt libraries for repeatable test generation, and training peers on AI-assisted workflow optimization
- Skilled in Agile/Scrum practices with hands-on JIRA board administration — configuring Kanban/Scrum boards, automation rules, custom fields, filters, and velocity/burndown reporting to optimize team delivery

TECHNICAL SKILLS

QA & Test Mgmt	JIRA (Workflows, JQL, Dashboards, Boards, Sprints, Filters, Automation Rules), XRay, Zephyr, TestRail, Confluence
Automation Tools	Selenium WebDriver, Pytest, Playwright, DeCAF, ACTS, RDK, Postman, Copycat
AI/ML Tools	Kiro AI, Claude (Anthropic), Amazon Q Developer, GitHub Copilot, Prompt Engineering, RAG Patterns
Languages	Python, JavaScript, SQL, HTML5, CSS3, Bash/Shell
CI/CD & DevOps	Jenkins, Git, GitHub Actions, Selenium Grid, Docker
Testing Types	Functional, Regression, Performance, System Integration, UAT, Smoke, Embedded/Firmware
Methodologies	Agile/Scrum, Kanban, TDD, BDD, SDLC, STLC, Risk-Based Testing
Platforms	Bluetooth LE, Wi-Fi, Zigbee, Thread, Matter, Android, RTOS/Linux, Salesforce

WORK EXPERIENCE

Amazon

05/2024 – Present

Senior Quality Assurance Lead

- Managed complete defect lifecycle in JIRA — creating custom JQL filters for defect fix-rate metrics, configuring dashboards for milestone tracking, and driving bug triage, WAR approval workflows, and launch readiness sign-off across multiple firmware releases
- Built automated JIRA reporting pipelines using JQL queries and JIRA REST APIs to generate defect trend analysis, per-component quality scorecards, and release retrospective metrics for executive stakeholders
- Developed Python-based automation frameworks (Pytest, DeCAF) for embedded firmware validation — covering functional, regression, and performance testing for FireTV and Alexa devices
- Leveraged AI tools (Kiro, Claude, Amazon Q) to accelerate QA workflows — AI-assisted test case generation, automated code review, intelligent log analysis, and self-healing test maintenance reducing manual effort by 40%
- Architected AI-powered release retrospective automation using prompt engineering and agent orchestration — auto-generating quality reports from JIRA data, Confluence pages, and TestRail metrics
- Designed and maintained JIRA sprint boards, Kanban workflows, and custom field configurations to track defect priority, severity classification, fix versions, and component-level quality health
- Optimized root cause analysis using AI log analyzers to transform system telemetry and ADB data into actionable insights, reducing mean-time-to-resolution for critical defects
- Built interactive QA dashboards (HTML/CSS/JS) with TestRail API and JIRA API integration, automating test execution reporting and release-readiness visualization
- Pioneered AI-driven test documentation workflows — using Claude and Kiro to auto-generate test plans, requirement traceability matrices, and release notes from JIRA epics, reducing documentation time by 60%
- Validated Salesforce-based internal applications through functional and regression testing, leveraging Copycat for automated UI test flows

Roku

04/2022 – 04/2024

Senior Quality Assurance Engineer

- Led development of comprehensive test strategies across TV and Audio platforms, integrating automated performance and regression testing into the firmware release lifecycle
- Managed JIRA sprint backlogs, Kanban boards, and defect workflows — creating custom JQL filters, automation rules, and dashboards to track fix rates, blockers, and release-readiness across multiple product lines
- Developed and optimized a modular Python automation library (RDK-based) for functional, regression, and performance testing with Git-based version control
- Integrated Python-based automation scripts into CI/CD pipelines; achieved 15% reduction in testing cycle time through automated reporting and JIRA workflow streamlining
- Configured JIRA sprint boards with velocity charts, burndown reporting, and custom fields to drive Agile delivery cadence and provide stakeholder visibility into quality health
- Spearheaded Post-Release Live Ops, monitoring real-time production telemetry and tracking post-release defects through JIRA to maintain system stability
- Adopted early AI-assisted debugging using GitHub Copilot and internal LLM tools to accelerate log parsing, identify failure patterns in firmware test results, and suggest root cause hypotheses — reducing triage time by 30%
- Integrated AI-powered test generation into sprint workflows — using generative AI to create regression test cases from JIRA user stories and acceptance criteria, increasing test coverage while reducing manual test authoring effort
- Built automated reporting dashboards integrating JIRA data with test execution results — verifying API responses, configuration propagation, and device behavior across platform variants

Google

10/2017 – 04/2022

Senior Quality Assurance Lead

- Led a team of QA engineers, defining test strategies and managing JIRA-based defect tracking, sprint planning, and backlog prioritization for Android-based IoT devices (smartwatches, Nest products)
- Developed advanced automation scripts using Python and ACTS framework to validate Bluetooth performance, Wi-Fi connectivity, and device-level behavior
- Managed defect workflows in JIRA — configuring issue type schemes, creating JQL-based filters for component-level defect analysis, and maintaining requirements traceability across releases
- Analyzed Bluetooth performance using Ellysis BT Sniffer, debugged logs with Wireshark, and conducted code flow analysis in time-critical environments
- Validated Salesforce platform: tested objects (standard/custom), layouts, field validations, sharing rules, permission sets, and data operations using Data Loader
- Developed Python-based test harnesses for automated device provisioning, OTA update validation, and multi-device interoperability testing across Google Home ecosystem
- Built custom data extraction scripts using SOQL queries and REST APIs to verify Salesforce data integrity; conducted code reviews enforcing automation best practices

SourceBits Technologies

12/2014 – 12/2015

Senior Software Engineer

- Orchestrated end-to-end mobile development lifecycle, translating business requirements into scalable architectural designs; achieved 30% increase in user engagement
- Developed and executed API tests using Postman; optimized complex SQL queries for large-scale data extraction and validation
- Engineered GitFlow strategy with automated merge-conflict detection and CI/CD quality gates, enabling continuous deployment across distributed teams
- Implemented unit and integration testing practices for Android applications, ensuring code quality and reducing regression defects across release cycles
- Collaborated with UI/UX architects to translate wireframes into pixel-perfect mobile features with responsive layouts across device form factors

MindTree

12/2012 – 11/2014

Senior Software Engineer

- Architected distributed Selenium Grid environment enabling simultaneous cross-browser execution, reducing test cycles by 60%
- Configured CI/CD build pipelines integrating automated test execution with Page Object Model design pattern for maintainable test suites
- Tracked and managed defects in JIRA through full lifecycle — from creation through triage, assignment, resolution, and verification — maintaining sprint reports and defect metrics
- Designed and executed modular testing strategies for complex subsystems, achieving 100% test coverage for mission-critical features with zero-defect leakage
- Contributed to high-velocity Agile/Scrum environments using JIRA boards for sprint planning, daily standups, and retrospectives
- Developed dynamic web applications implementing AJAX and RESTful API services; built reusable AngularJS components for data visualization

Technical Associate

- Tested cross-browser compatibility using Selenium WebDriver with XPath and CSS selectors for dynamic element identification; logged and tracked defects in JIRA with detailed reproduction steps, severity classification, and screenshots
- Created and executed manual and automated test cases for web applications, covering functional, regression, and smoke testing across multiple browser environments (IE, Firefox, Chrome)
- Participated in all SDLC/STLC phases from requirements gathering through production deployment; contributed to test case design, test plan documentation, and unit testing guidelines
- Developed responsive web interfaces using JavaScript/jQuery with HTML5/CSS3, implementing cross-platform compatibility and WCAG accessibility standards
- Built interactive UI components and implemented AJAX-based asynchronous server communication, improving page responsiveness and reducing full-page reload dependencies
- Participated in daily Agile stand-ups, sprint planning, and defect triage meetings — using JIRA to track task progress, estimate story points, and report sprint velocity
- Performed database validation using SQL queries to verify data integrity between frontend interfaces and backend systems; validated CRUD operations and data flow across application layers
- Documented test results, defect reports, and release notes; maintained traceability matrices linking requirements to test cases for audit compliance

EDUCATION

Masters in Information Systems (MIS) — University of Illinois at Springfield • GPA: 3.8/4.0

MBA in Project Management — Sikkim Manipal University • GPA: 3.9/4.0

Bachelor of Engineering in Computer Science — KS Institute of Technology • GPA: 3.6/4.0

CERTIFICATIONS

- Prompt Engineering Certification — Master AI
- Salesforce Certified Administrator (SCA)
- Python for Everybody Specialization
- Certified Scrum Developer
- Business Intelligence — University of Illinois Springfield
- IT Project Management — University of Illinois Springfield